

W-3. Water Harvesting Reservoir with Dam Construction				
Water Resources Development	Medium	High		
Measure Type	Cost	Preferences by Community	Hazards	Implementation Time of Measures

Descriptions									
Measures	Concern regarding water scarcity is developing globally, especially in areas that are vulnerable to drought and erratic rainfall patterns. Constructing water harvesting reservoirs, frequently combined with dam development, provides a workable way to efficiently manage water supplies. By collecting extra rainfall, these reservoirs can provide a continuous supply of water for residence, commercial, and agricultural applications. However, sustainability and the effects on the environment must be carefully considered when developing such infrastructures.								
	The construction of water harvesting reservoirs with dam facilities in the CHT is a critical strategy for addressing water scarcity and ensuring sustainable resource management. Through the implementation of comprehensive conservation measures, including community involvement, erosion control, sustainable design practices, detailed site evaluations, and effective monitoring, we can develop resilient water systems that not only support human needs but also save the region's rich biodiversity. Communities in the CHT will benefit from these initiatives, which aim to maintain a harmonious balance between environmental preservation and the development of water resources.								
Location	R-114	R-99	R-65	K-98	K-76	K-71	B-173	B-77	B-76
	■	■	■	■	■	■	■	■	■
Applicable Sites	• Upland Catchment Areas • Seasonal Riverbeds & Streams • Gently Sloping Valleys • Deforested or Degraded Land								
Slope	Less than 3%		3% to 15%		15% to 30%		30% to 60%		Above 60%
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Design options and performance	• Stores rainwater and runoff, suitable for seasonal streams. • Check Dams & Percolation Ponds Slows water flow, promotes groundwater recharge. • Terraced Reservoirs – Built in steep terrains, reducing soil erosion.								
Feasibility criteria	Technical Feasibility • Ensures soil and slope stability to prevent dam failure. • Evaluates rainfall patterns and runoff potential. • Determines storage needs and distribution efficiency. Environmental Feasibility								

	• Reduces erosion and sedimentation. • Avoids disrupting aquatic and terrestrial habitats. • Prevents excessive evaporation and over-extraction. Economic Feasibility • Balances construction and maintenance costs with long-term benefits. • Provides water for agriculture, livestock, and domestic use. • Uses local materials and community-led maintenance. Social Feasibility • Engages local populations in construction and management. • Ensures safe drinking water, sanitation, and irrigation support. • Aligns with government policies and funding programs.			
Operation and maintenance	• Routine Maintenance – Regular desilting, embankment inspection, and vegetation control. • Structural Repairs – Ensures dam stability, leak prevention, and overflow management. • Seasonal Adjustments – Prepares for monsoon flooding and dry-season water conservation. • Community Involvement – Establishes local water user groups for monitoring and upkeep.			
Cost	Medium			
Benefits	Benefits are as follows-			
	Type	Low	Medium	High
	Canopy Cover Increase			
	Carbon Sequestration			
	Crop Production Increase			
	Crop Failure Risk Reduction			
	Ensuring Food Security			
	Fruit Production Increase			
	Forestry Goods Production Increase			
	Livestock Production Increase			
	Fisheries Production Increase			
	Alternative Livelihood Generation			
	Increase of Income			
	Soil Erosion Control			
	Improved Soil Health			
	Soil Fertility Increase			
	Sustainable Water Supply			
	Water Quality Improvement			
	Water Purification			
	River Bank Erosion Control			
	Surface Runoff Reduction			
	Groundwater Recharge			

	Pest and Disease Management			
	Reduction of Pesticides and Fertilizer Use			
	Ecosystem and Biodiversity Increase			
	Better Communications			
	Better Education			
	Better WASH facilities			
Implementation Challenges				
	Type	Low	Medium	High
	Land unavailable/high land area required			
	Supply chain disconnection			
	Land right related problems			
	High human investment capital need			
	Insufficient infrastructures			
	Less demand of agroforestry product			
	Environmentally unsustainable			
	Socially unacceptable			
	Less beneficial			
Environmental performance	• Water Conservation – Enhances groundwater recharge and reduces wastage. • Climate Resilience – Mitigates drought effects and regulates seasonal water availability. • Ecosystem Protection – Supports wetland restoration and biodiversity conservation.			
Remarks	Selecting the right site is critical, as locations must be geologically stable and hydrologically viable to ensure long-term functionality. The sustainability of these projects depends on regular maintenance, with active community participation playing a key role in their success. Additionally, reservoirs should be integrated into broader watershed management efforts to enhance overall catchment conservation. Collaboration between the government and NGOs is essential, providing policy support, funding, and technical expertise to ensure efficient implementation and long-term viability.			